

Solution

LIGHT -01

Class 10 - Science

1. (a) away from principal axis

Explanation: If these parallel beams non-parallel to the principal axis fall on the convex lens, one may pass through the optic centre and pass without any deviation. another may pass through focus and after refraction passes parallel to the principal axis. so they converge at a point away from the principal axis.

2. (a) 5 m

Explanation: New distance between the man and the mirror = $y + 5$

$\therefore$ The distance between the man and his image

$$(y + 5) \times 2 = 20$$

$$2y + 10 = 20$$

$$2y = 10$$

$$y = 5 \text{ m}$$

- 3.

(b) will emerge parallel to the principal axis

Explanation: The ray light passing through the principal focus of the convex lens will emerge as parallel to the principal axis after refraction from the convex lens.

- 4.

(c) is less than one

Explanation: The convex mirror is used in the rearview mirror. The convex mirror always gives a smaller image. Hence, magnification produced by the rearview mirror is always less than 1.

5. (a) greater than the angle of incidence

Explanation: greater than the angle of incidence

- 6.

(b) a scale and a screen

Explanation: Screen for image formation and scale to measure length are required.

- 7.

(d) Black absorbs all the colours

Explanation: Black absorbs all the colours

- 8.

(b) Q to R

Explanation: Q to R

- 9.

(d) it is refracted away from the normal

Explanation: it is refracted away from the normal

- 10.

(b) three different points

Explanation: Red, blue, and green lights have different wavelengths so they will be refracted accordingly. So three points of convergence on the principal axis exist.

11. (a) 2.5 D

Explanation:

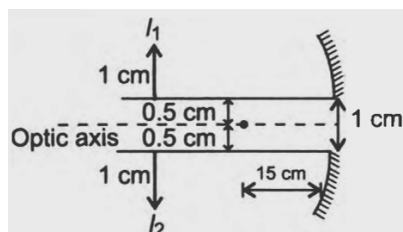
The power of a lens is the reciprocal of its focal length. It is a measure of the degree of convergence or divergence of light rays achieved by a particular lens. The power of a convex lens is positive.

$$\text{Power} = \frac{1}{\text{focal length}}$$

Given, focal length = 40 cm or 0.40 m

$$\text{Therefore, Power of the given convex lens} = \frac{1}{\text{focal length}} = \frac{1}{0.40} = 2.5 \text{ D}$$

12.
(c) Zero
Explanation: Power of a plane lens is zero.
13.
(b) ∞
Explanation: Image of an object at $2f$ will be at $2f$, while it will be at ∞ , if object is at f .
14.
(c) 90 degrees
Explanation: 90 degrees
15.
(d) 0 degree
Explanation: 0 degree
16.
(c) smaller than the angle of incidence
Explanation: smaller than the angle of incidence
17.
(b) A convex lens of focal length 5 cm
Explanation:
A convex lens (converging lens) of focal length 5 cm should be used to read small letters in a dictionary. A convex lens of short focal length (5 cm) bends the light rays through large angles, by focussing them closer to the optical centre.
18.
(c) 3
Explanation: 3
19.
(b) at the centre of curvature
Explanation: at the centre of curvature
20.
(d) 1,1
Explanation: We know, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$
Here, $f = -10$ cm, $u = -15$ cm
 $\therefore \frac{1}{-10} = \frac{1}{-15} + \frac{1}{v}$
or, $\frac{1}{-10} = \frac{1}{-15} + \frac{1}{v} = -\frac{1}{30}$
or, $v = -30$ cm
Hence, two images will be formed at a distance of 30 cm from mirror.
We know, $m = -\frac{v}{u} = -\frac{(-30)}{(-15)} = -2$
Distance between broken mirrors is 1 cm
 $h_{i1} = mh_{o1} = (-2) \times (-0.5) = 1$ cm
Image (I_1) due to upper part of the mirror will be formed at 1 cm above its optic axis.
Similarly, image (I_2) due to lower part of the mirror will be formed at 1 cm below its optic axis



21.
(b) The speed of light rays decreases when they enter the converging lens.
Explanation: The speed of light decreases while travelling from rarer to denser medium.

22. (b) more than 200 mm
Explanation: Since the image formed is upright, the object lies within the focus of the concave mirror.
 So, $f > 100 \text{ mm}$
 We know $R = 2f$
 So, $R > 200 \text{ mm}$
 i.e. Radius of curvature is more than 200 mm.
23. (b) on the same walls as the window
Explanation: Since reflection of light received from the distant object can be received only then.
24. (b) 15cm
Explanation: 15cm
25. (d) $u = -1.05 \text{ cm}$ and $m = 19.05 \text{ cm}$
Explanation: $u = -1.05 \text{ cm}$ and $m = 19.05 \text{ cm}$
26. (a) Real, magnified, inverted
Explanation: The image formed is real, magnified and inverted because projector is based on this kind of image formation. In this case object is placed in between F and 2F.
-
27. (d) 16 cm
Explanation: Radius of curvature, $R = 32 \text{ cm}$
 Focal length, $f = ?$
 We know that,
 $f = \frac{R}{2}$
 $f = \frac{32}{2} = 16 \text{ cm}$
28. (a) towards the screen
Explanation: The blurred image is seen only when the separation between the screen and mirror is less than the focal length.
29. (d) the foots are in straight line
Explanation: To avoid any change due to bend needles, the feet are to be arranged in a straight line.
30. (b) at a distance greater than radius of curvature
Explanation: at a distance greater than radius of curvature
31. (b) S
Explanation: S
32. (a) the mirror and the screen
Explanation: The distance between the mirror and the screen will give the focal length of the mirror as the mirror focuses the light on the screen.
33. (d) -, +
Explanation: -, +
34. (d) 6 cm
Explanation: $u = -15 \text{ cm}$, $v = -10 \text{ cm}$

$$f = ?$$

We know that,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{(-10)} + \frac{1}{(-15)} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{f} = -\frac{1}{10} - \frac{1}{15} = \frac{-3-2}{30} = -\frac{5}{30} = -\frac{1}{6}$$

$$\therefore f = -6 \text{ cm}$$

$\therefore$ The focal length of the concave mirror is 6 cm.

35.

(c) In (a), the incident rays are diverged after refractions, so lens is concave.

Explanation: In (a), the incident rays are diverged after refraction, so the lens is concave and in (b), the incident rays are converged after refraction, so the lens is convex.

36.

(c) Real and Inverted

Explanation: Real and Inverted

37.

(d) material C

Explanation: material C

38.

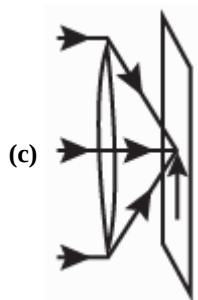
(b) 1.6244

Explanation: 1.6244

39. **(a)** increase

Explanation: According to Snell's law, ratio of the sine of the angle of incidence to the sine of the angle of refraction is always constant for a given pair of media. Therefore, if the angle of incidence increases, the angle of refraction also increases proportionally to the increase of incidence.

40.



Explanation: Here the image formed should be real and inverted, since parallel rays coming from infinity after passing through the lens converge to the focus and the image formed is real and inverted.

41.

(d) 7.5 m

Explanation: The reason being, the image formed by a plane mirror is the same distance behind the mirror as it is between the object and the mirror. A distance of 5 m between man and his image means that the distance between him and the mirror = $\frac{5}{2} = 2.5$ m

Thus, the distance he should walk = $10 - 2.5 = 7.5$ m

42. **(a)** Between F and 2F

Explanation: When an object is placed at a distance more than twice the focal length (F) from a concave mirror, the image will be formed at a distance between F and 2F.

43. **(a)** substance S

Explanation: substance S

44.

(c) does not get refracted

Explanation: does not get refracted

45. **(d)** Total internal reflection
Explanation: Total internal reflection
46. **(a)** $i = r = 90^\circ$
Explanation: It is because when i is 90 degrees, it means incident ray is perpendicular to the refracting surface, and light travels in the shortest path that's why it bends towards the normal when it enters a denser medium. But we know that the shortest distance is perpendicular to the medium. So refracted ray doesn't bend and continues to move straight.
47. **(b)** 3 m
Explanation: 3 m
48. **(a)** Less than 10 cm
Explanation: Only when the object is positioned between the pole and the focus can a concave mirror produce an upright image.
Distance between pole and focus = Focal length = 10 cm
As a result, the object can be placed less than 10 cm from the mirror. As a result, the image created when an object is put between the pole and the focus. The virtual, upright, and expanded concave mirror.
49. **(b)** is refracted away from the normal
Explanation: is refracted away from the normal
50. **(d)** 0 degree
Explanation: 0 degree
51. **(d)** sun
Explanation: Larger the distance, better and sharper focus can be achieved.
52. **(d)** increase
Explanation: Since $n = \frac{\sin i}{\sin r} = \text{constant}$, with an increase in i , r will increase.
53. **(c)** Both concave
Explanation:
The mirror and the lens, both are concave. The distance of the principal focus measured from the pole is called the focal length. According to the New Cartesian Sign Convention, the focal length of a concave mirror as well as a concave lens is considered negative.
54. **(a)** a bulging-out surface
Explanation: a bulging-out surface
55. **(a)** virtual
Explanation: virtual
56. **(d)** Between the pole of the mirror and its principal focus.
Explanation: When an object is placed between the pole and the principal focus of a concave mirror, a virtual and erect, and an enlarged image is formed behind the mirror.
57. **(b)** 0.75
Explanation: 0.75
58. **(b)** very near to the focus of the reflector.
Explanation: In torches, search lights and headlights of vehicles concave mirrors are used. When a light source is placed at the

focus of the mirror, after reflection light rays become parallel to the principal axis. Due to this, the bulb is placed very near to the focus of the reflector.

59.

(d) 15 cm

Explanation: We know, $m = \frac{h_i}{h_o} = \frac{-v}{u}$

Here, $m = 3$, or, $-m = \frac{v}{u}$

So, $m = \frac{-v}{u}$ or, $-m = \frac{v}{u}$

or $-m - 1 = \frac{v}{u} - 1$ or $-m - 1 = \frac{v-u}{u}$

or $-3 - 1 = \frac{40}{u}$ or $-4 = \frac{40}{u}$

or $u = -10$ cm

$\therefore v - u = 40$ cm

or $v + 10 = 40$ cm

We know, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

So $f = \frac{uv}{u+v}$

or $f = \frac{-300}{20} = -15$ cm

Thus, magnitude of focal length of the mirror is 15 cm.

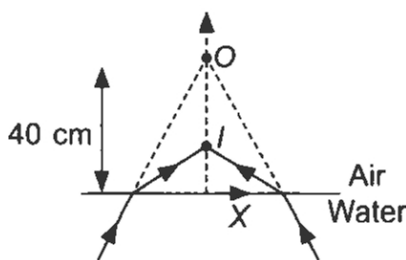
60.

(c) Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.

Explanation: When we cover the half portion of the convex lens with black paper, the full image of the object can be seen but the intensity of image is reduced. Rays incident on convex lens are refracted from uncovered part only and form the full image.

61. (a) 30 cm

Explanation: A ray of light travels from water across the interface through air and converges to form an image at I. Here, medium 1 is water and medium 2 is air.



$d_1 = +40$ cm, $\mu_1 = \frac{4}{3}$, $\mu_2 = 1$

As, $\frac{\mu_2}{\mu_1} = \frac{\text{Real depth } (d_2)}{\text{Apparent depth } (d_1)}$

So, $\frac{\mu_1}{d_1} = \frac{\mu_2}{d_2}$ or $d_2 = \frac{\mu_2}{\mu_1} d_1$,

we get $d_2 = +30$ cm

Thus, the image is formed 30 cm above the interface.

62.

(c) all reflecting surfaces

Explanation: The laws of reflection hold true for all the reflecting surfaces either spherical or plane. Therefore, the plane mirror, concave mirror, and the convex mirror all will follow the laws of reflection.

63.

(d) $P_1 + P_2$

Explanation: The net power of the lenses placed in contact is given by the algebraic powers of the individual powers.

Net power of the lens combination, $P = P_1 + P_2$

64. (a) More than 16 metres

Explanation: The image of R is 8 m behind the mirror (say R') perpendicularly. However, with respect to camera at M, it is the diagonal distance (MR'), which is more than 16 m.

65. (a) $\frac{3}{2}$

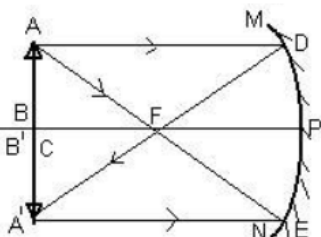
Explanation: If c is the speed of light in vacuum, and v is the speed of light in the medium, then refractive index of the medium n_m is given by:

$$n_m = \frac{\text{speed of light in vacuum}}{\text{speed of light in the medium}} = \frac{c}{v}$$

$$\therefore n_m = \frac{c}{v} = \frac{3}{2}$$

66. (a) 30 cm in front of the mirror

Explanation:



The ray diagram illustrates that, in the case of a concave mirror, the image size becomes equal to the object size when the object distance equals the radius of curvature, i.e., twice the focal length.

So, the object has to be placed at a distance of $(15 \times 2) = 30$ cm in front of the concave mirror.

67. (d) Twinkling of stars
Explanation: Twinkling of stars is due to atmospheric refraction. Distant star acts like a point source of light. When the starlight enters the earth's atmosphere it undergoes refraction continuously, due to changing refractive index i.e. from Rarer to denser medium. It bends towards the normal successively, hence the amount of light enters our eyes fluctuates sometime bright and sometime faint.
68. (a) At 2F
Explanation: When an object is placed at 2F in front of a convex lens, the image is also formed at 2F. The image is real and inverted and is of the same size as that of the object.
69. (d) all mirrors irrespective of their shape
Explanation: We know that from the laws of reflection, the incident ray, the reflected ray, and the normal to the reflecting surface all lie in the same plane. Also, the angle of reflection is equal to the angle of incidence.
 The laws of reflection hold good for all reflecting surfaces irrespective of their shapes whether plane or curved.
70. (d) $R = 2f$
Explanation: $R = 2f$
71. (d) Bends towards the normal
Explanation: Speed of light changes when it travels from one medium to another. The speed of light decreases while it travels from rarer to denser medium. Hence it bends towards normal. On the other hand, the speed of light increases while it travels from denser to rarer medium. Hence, it bends away from normal.
72. (d) depends on the refractive index of the medium
Explanation: The refractive index affects the emergent ray. Hence, it depends on the refractive index of the medium.
73. (a) between pole and focus
Explanation: between pole and focus
74. (b) f
Explanation: Since the lens is cut into two equal parts without affecting its curvature, it means the radius of curvature R is same for both parts and hence the focal length ($F = \frac{R}{2}$) will remain same.
75. (c) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
Explanation: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$